

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250282251

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

YANG; Seong Yeol

CHARGING MANAGEMENT DEVICE, WIRELESS CHARGING SYSTEM, SERVER, AND METHOD FOR PROVIDING WIRELESS CHARGING SERVICES

Abstract

Discussed is a wireless charging system including a wireless charging station installed under the ground on a road and configured to wirelessly transmit power within a predetermined region of the road, and a charge control device configured to control charging of the wireless charging station, wherein the charge control device includes a communication unit configured to communicate with a server to receive whether a power transmission rate for a location of a vehicle when the vehicle is stopped at a chargeable position of the wireless charging station is equal to or greater than a reference value, and a control unit configured to control an operation state of the wireless charging station based on vehicle information and whether the power transmission rate is equal to or greater than the reference value received from the server.

Inventors:	YANG; Seong Yeol (Daejeon, KR)
Applicant:	LG ENERGY SOLUTION, LTD. (Seoul, KR)
Family ID:	1000008626874
Assignee:	LG ENERGY SOLUTION, LTD. (Seoul, KR)
Appl. No.:	19/217547
Filed:	May 23, 2025

Foreign Application Priority Data

KR	10-2019-0092312	Jul. 30, 2019
----	-----------------	---------------

Related U.S. Application Data

Publication Classification

Int. Cl.: **B60L53/66** (20190101); **B60L53/12** (20190101); **B60L53/30** (20190101); **G06Q30/04** (20120101); **H02J50/40** (20160101); **H02J50/80** (20160101); **H02J50/90** (20160101)

U.S. Cl.:

CPC **B60L53/665** (20190201); **B60L53/12** (20190201); **B60L53/305** (20190201); **G06Q30/04** (20130101); **H02J50/40** (20160201); **H02J50/80** (20160201); **H02J50/90** (20160201); **B60L2240/62** (20130101); **B60L2260/22** (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Divisional of U.S. patent application Ser. No. 17/631,248, filed on Jan. 28, 2022, which was filed as the National Phase of PCT International Application No. PCT/KR2020/010094, filed on Jul. 30, 2020, which claims priority to Korean Patent Application No. 10-2019-0092312, filed in the Republic of Korea on Jul. 30, 2019, the entire contents of which is hereby expressly incorporated by reference into the present application.

BACKGROUND

Field

[0002] The present disclosure relates to a charging management device performing scheduling for charging a vehicle while being at a standstill for waiting for a signal in traveling, a wireless charging system, a server, and a method for providing wireless charging services.

Discussion of the Related Art

[0003] In recent years, research and development for secondary batteries have been actively conducted. Here, the secondary battery is a battery capable of recharging and discharging, and in its meaning, includes all of the existing Ni/Cd battery, the Ni/MH battery, and the like and a recent lithium-ion battery. Among secondary batteries, lithium-ion batteries have the advantage of having much higher energy density than the existing Ni/Cd batteries and Ni/MH batteries. In addition, lithium-ion batteries may be manufactured in small size and light weight, and accordingly, may be used as power sources for mobile devices. The lithium-ion batteries are drawing attention as a next-generation energy storage medium as their range of use has expanded to a power source for electric vehicles.

[0004] The secondary battery is used as a battery pack that includes a battery module in which a plurality of battery cells are connected in series and/or parallel, and a battery management system (BMS) that controls the operation of the battery module and manages the state of the battery module.

[0005] In general, electric vehicles require a lot of energy to improve mileage, and thus battery cells are connected in parallel to increase capacity. Accordingly, shortening the charging time of the battery of the electric vehicle has emerged as an important problem. However, in the related art, there are limitations in that charging of the electric vehicle is possible only when the vehicle is parked in a specific place such as an electric charging station, and it is needed to move to a place where a charger is installed even when charging is urgently required.

SUMMARY OF THE DISCLOSURE

[0006] The present disclosure has been made in order to solve the problems described above, and an object thereof is to provide a system and service capable of charging a vehicle that is waiting for a signal in traveling and efficiently managing vehicle charging by scheduling charging time to a destination.

[0007] According to an aspect of embodiments of the present disclosure, there is provided a charging management device managing charging of a vehicle battery, including at least one processor, at least one memory that stores computer program instructions that, when executed, cause the at least one processor to perform operations, a communication device configured to communicate with a server, and a power receiving device configured to receive power wirelessly, wherein the operations include searching for a vehicle traveling route at least based on a destination of a vehicle, identifying wireless charging stations available to make charging while being stopped based on the traveling route, and receiving power from the wireless charging station when stopped in charging available positions at intersections in which the identified wireless charging stations are installed.

[0008] In embodiments of the present disclosure, a list of the charge-available wireless charging stations may be received from the server in response to transmitting the traveling route to the server.

[0009] In embodiments of the present disclosure, the operations may further include determining a required amount of charge required for charging and transmitting the determined required amount of charge to the server.

[0010] In embodiments of the present disclosure, the operations may further include, if power is received from the wireless charging station, updating the required amount of charge based on the received power, and transmitting the updated required amount of charge to the server.

[0011] In embodiments of the present disclosure, the operations may further include calculating an estimated arrival time to the charge-available wireless charging station, and transmitting the calculated estimated arrival time to the server.

[0012] In embodiments of the present disclosure, the operations may further include transmitting a location of the vehicle to the server.

[0013] According to another aspect of embodiments of the present disclosure, there is provided a wireless charging system including a wireless charging station installed under the ground of a road and configured to wirelessly transmit power within a predetermined region, and a charge control device configured to control charging of the wireless charging station, wherein the charge control device includes a communication unit configured to communicate with a server, and a control unit configured to control an operating state of the wireless charging station based on vehicle information received from the server and signal information of traffic lights.

[0014] In embodiments of the present disclosure, the vehicle information may include identification information for identifying the vehicle and location information indicating a location of the vehicle.

[0015] In embodiments of the present disclosure, the wireless charging station may include a vehicle location determination device configured to determine whether the vehicle is stopped within the predetermined region, a wireless power transmission unit configured to wirelessly transmit power to the vehicle, and a power amount calculation unit configured to calculate an amount of power transmitted by the wireless power transmission unit.

[0016] In embodiments of the present disclosure, the charge control device may be configured to receive the amount of power for charging the vehicle from the wireless charging station, and transmit, to the server, the received amount of power in association with the vehicle information.

[0017] In embodiments of the present disclosure, the vehicle location determination device may be configured to determine whether the vehicle is stopped at a location in which a power transmission rate is equal to or greater than a reference value if power is wirelessly transmitted to the vehicle.

[0018] In embodiments of the present disclosure, the charge control device may be configured to

control an operation state of the wireless charging station to switch from a sleep state to a standby state when the signal information of the traffic lights indicates that the vehicle is to be stopped in a traveling direction.

[0019] In embodiments of the present disclosure, the charge control device may be configured to control the wireless charging station to stop power transmission before a signal for the traveling direction of the vehicle in the traffic lights is changed from a stop signal to a go signal.

[0020] According to another aspect of embodiments of the present disclosure, there is provided a server for managing wireless charging of a vehicle on a road, including at least one processor, at least one memory that stores computer program instructions that, when executed, cause the at least one processor to perform operations, and a communication device configured to communicate with the vehicle and a charge control device, wherein the operations include receiving, from the vehicle, a traveling route and location information of the vehicle, identifying at least one wireless charging station through which the vehicle passes on the traveling route based on the traveling route and the location information, and providing information on at least one identified wireless charging station to the vehicle.

[0021] In embodiments of the present disclosure, the operations may further include transmitting information on the vehicle to a corresponding charge control device that controls the at least one identified wireless charging station.

[0022] In embodiments of the present disclosure, the operations may further include receiving, from the charge control device, an amount of power charged in the vehicle in association with the information on the vehicle, and generating billing information for the vehicle based on the received amount of power.

[0023] In embodiments of the present disclosure, the generating of the billing information may be performed based on a sum of all the amount of power received by the vehicle on the traveling route.

[0024] According to another aspect of embodiments of the present disclosure, there is provided a method for providing wireless charging services for wirelessly charging a vehicle while the vehicle is at a standstill by using a plurality of wireless charging stations provided under the ground of a road, including confirming, at a server, at least one wireless charging station available to make charging, among the plurality of wireless charging stations, based on a traveling route of the vehicle, transmitting power from the wireless charging station to the vehicle if the vehicle is stopped in a charging available position at an intersection in which the at least one wireless charging station is installed, and generating billing information based on an amount of power transmitted to the vehicle.

[0025] In embodiments of the present disclosure, the method may further include receiving payment method information from the vehicle, and transmitting payment information to a payment server based on the payment method information and the billing information.

[0026] In embodiments of the present disclosure, the billing information may be generated based on vehicle information recognized by identifying a license plate of the vehicle.

[0027] With the configuration described above, the system and service according to the embodiments of the present disclosure may charge a vehicle that is waiting for a signal in traveling and may efficiently manage vehicle charging by scheduling charging time to a destination.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] FIG. 1 is a diagram schematically illustrating a situation in which the present disclosure is applied.

[0029] FIG. 2 is a diagram illustrating a configuration of a system according to embodiments of the

present disclosure.

[0030] FIG. 3 is a diagram illustrating a situation in which a system according to embodiments of the present disclosure operates.

[0031] FIG. 4 is a block diagram illustrating a functional configuration of a charging management device according to embodiments of the present disclosure.

[0032] FIG. 5 is a block diagram illustrating a hardware configuration of the charging management device according to embodiments of the present disclosure.

[0033] FIG. 6 is a flowchart illustrating an operation of the charging management device according to embodiments of the present disclosure.

[0034] FIG. 7 is a block diagram illustrating a functional configuration of a server according to embodiments of the present disclosure.

[0035] FIG. 8 is a block diagram illustrating a hardware configuration of the server according to embodiments of the present disclosure.

[0036] FIG. 9 is a flowchart illustrating an operation of the server according to embodiments of the present disclosure.

[0037] FIG. 10 is a block diagram illustrating a functional configuration of a wireless charging system according to embodiments of the present disclosure.

[0038] FIG. 11 is a flowchart illustrating an operation of the wireless charging system according to embodiments of the present disclosure.

[0039] FIG. 12 is a flowchart illustrating a method for calculating an available charging time of a vehicle by the wireless charging system according to embodiments of the present disclosure.

[0040] FIG. 13 is a block diagram illustrating another example of a functional configuration of a charge control device according to embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0041] Hereinafter, various embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. In the present disclosure, the same reference numerals are used for the same elements in the drawings, and duplicate descriptions for the same elements are omitted.

[0042] In various embodiments of the present disclosure disclosed herein, specific structural or functional descriptions are merely by example for the purpose of describing embodiments of the present disclosure, and various embodiments of the present disclosure may be implemented in various forms and should not be construed as being limited to the embodiments described herein.

[0043] As used in various embodiments, the terms such as “first”, “second”, “the first”, or “the second” may modify various components, regardless of order and/or importance, but do not limit the components. For example, without departing from the scope of the present disclosure, a first element could be termed a second element, and similarly, in reverse, a second element could be termed a first element.

[0044] The terms and phrases as used herein are merely provided to describe specific embodiments, and may not be intended to limit the scope of other embodiments. A singular form is intended to include a plural form, unless the context clearly indicates otherwise.

[0045] FIG. 1 is a diagram schematically illustrating a situation in which the present disclosure is applied.

[0046] A vehicle stops to wait for the go signal if a signal of traffic lights related to a traveling direction at an intersection or the like is the stop signal. The length of the stop signal and the like of the traffic lights is determined by a signal controller controlling each traffic light. Therefore, it is possible to calculate the approximate time for the vehicle to be at a standstill at the intersection by the stop signal. Embodiments of the present disclosure are made based on the above idea.

[0047] In the system and service according to embodiments of the present disclosure, a wireless charging station capable of wirelessly charging a vehicle's battery may be installed in a region or a location where the vehicle is stopped in order to wait for a signal at an intersection. If the vehicle is

completely stopped within the chargeable region, wireless charging may be performed by the request of a user (driver) or according to a vehicle setting. The period in which the stop signal and the go signal are repeated is predetermined for the signals of the traffic lights at each intersection. Therefore, if a route to the destination of the vehicle is derived, the signal period at each intersection on the derived route may be used to calculate at which point the vehicle will stop for waiting for a signal, how long it will be at a standstill, and so on. As a result, it is possible to calculate the total charging time to be performed while the vehicle is traveling to its destination. Here, the stop signal may be a concept including a signal other than the go signal in a direction in which the vehicle is traveling. For example, if the vehicle intends to go straight at an intersection, a signal indicating that the vehicle should not go in the traveling direction, such as a left turn signal, may be included.

[0048] Referring to FIG. 1, a traveling route for a vehicle **100** to move to a destination is indicated by a thick arrow. The traveling route may be a result of searching by a navigation provided in the vehicle **100**, a navigation application installed on a user device (smartphone, tablet, or the like) possessed by a user of the vehicle **100**, and the like, based on the destination input by the user. Alternatively, the traveling route may be a result of searching by the server based on information related to the current location and destination provided from the vehicle **100** to a server such as a charging management server.

[0049] On the traveling route, there are a plurality of intersections through which the vehicle **100** will pass. At each intersection, a chargeable region capable of wireless charging is provided in a predetermined region or location. In some cases, there may be no vehicle waiting for a signal in the chargeable region. Depending on the traffic conditions, there may be many moving vehicles, and other vehicles may be at a standstill in some or the entire chargeable region. Accordingly, charging may be performed if the vehicle **100** stops in the chargeable region and power transmission is possible from the wireless charging station installed in the chargeable region to the vehicle **100** at a power transmission rate equal to or greater than the reference value.

[0050] The vehicle **100** may be an electric vehicle. Therefore, the vehicle **100** may use a motor to generate power, and may include a battery for driving the motor. The battery may be a secondary battery capable of charging and discharging, and a sufficient amount of power is to be charged for the vehicle **100** to move to the destination. Depending on the state of charge (SOC) of the battery provided in the vehicle **100**, the battery of vehicle **100** may be required to be charged in traveling. In some cases, if the battery is not in a fully charged state, it may be desired to always charge the battery of the vehicle **100** just in case. In other cases, it may be desired to charge the battery of the vehicle **100** only in the minimum amount required to its destination in terms of charging cost. For example, if the destination is home or a place where charging is possible at a lower price than a wireless charging station, such as inexpensive charging stations, it may be desired to perform charging only in the minimum amount.

[0051] When the vehicle **100** is waiting for a signal by the stop signal or the like at an intersection, determination is made as to whether or not it is in a chargeable state. The chargeable state may be, for example, a state in which the power transmission rate is equal to or greater than a reference value. That is, the chargeable state may be a state in which wireless charging efficiency is equal to or greater than a reference value. The wireless charging efficiency may be determined by various factors such as a relative location between a location where the vehicle **100** is stopped and the charging region, a condition of the battery installed in the vehicle **100**, weather, and surrounding environment.

[0052] When the vehicle **100** is in a chargeable state, charging is performed if a predetermined condition is met. The predetermined condition may be a case where a charging execution request is transmitted from the vehicle **100** to a wireless charging station, a charge control device including the wireless charging station, or a server that manages charging. The charging execution request may be transmitted by a user to a related device (such as a server described above) through the

vehicle **100** or a user device immediately before or during traveling. Alternatively, the charging execution request may be automatically transmitted whenever the user starts traveling by subscribing to the wireless charging service in advance. Alternatively, the charging execution request may be transmitted by the user to a wireless charging station installed at an intersection or a charge control device including the wireless charging station when the vehicle **100** stops at the intersection. The transmission method for the charging execution request is by example and is not limited thereto.

[0053] In addition, before starting charging in response to the charging request signal, a request for charging execution confirmation may be transmitted to the user. Charging may be started only if the user accepts the request for charging execution confirmation and instructs that charging should be executed. However, this is illustrative, and if the charging execution request is transmitted by the user, and the condition that the vehicle **100** should be stopped at a location of the chargeable region having a wireless charging efficiency equal to or greater than the reference value is satisfied, charging may be automatically performed.

[0054] As described above, the vehicle **100** may be put into a situation of waiting for a signal in traveling to a destination, and if the stop region during signal waiting corresponds to a chargeable region, the battery may be charged at any time during the signal waiting stop.

[0055] Hereinafter, a method for performing wireless charging of the vehicle **100** in a chargeable region will be described in detail. In other words, a method for providing a wireless charging service to the vehicle **100** will be described.

[0056] FIG. **2** is a diagram illustrating a configuration of a system according to embodiments of the present disclosure.

[0057] A system according to embodiments of the present disclosure may include the vehicle **100**, a server **200**, and a wireless charging system **300**.

[0058] The vehicle **100** may be an electric vehicle that moves by driving a motor with a battery, as described above in FIG. **1**. The battery is a rechargeable secondary battery. The vehicle **100** may include a wireless charging circuit in order to wirelessly charge the mounted battery.

[0059] The vehicle **100** may include a charging management device that manages charging of the battery. The charging management device searches for a vehicle traveling route at least based on a destination of the vehicle, and confirms wireless charging stations **320** that are charge-available while being at a standstill based on the traveling route. The charging management device may receive power from the wireless charging stations **320** if the vehicle is stopped in chargeable locations at intersections in which the confirmed wireless charging stations **320** are installed.

[0060] The vehicle **100** may communicate with the server **200** and the wireless charging system **300** through a network **500**. The vehicle **100** may be connected to the network **500** using an in-vehicle connectivity system. Alternatively, the vehicle **100** may be connected to the network **500** through a communication unit provided separately from the in-vehicle connectivity system. In addition, the vehicle **100** may be connected to the network **500** in a method other than a method using a communication unit of the vehicle itself. For example, the vehicle **100** may be connected to the network **500** through the mobile device of the user. Alternatively, instead of the vehicle **100** being connected to the network **500**, the mobile device of the user may be connected to the network **500** to perform various operations to be described below.

[0061] The charging management device of the vehicle **100** may receive a list of charge-available wireless charging stations **320** from the server **200** in response to transmitting the traveling route to the server **200**. Additionally, the charging management device of the vehicle **100** may determine a required amount of charge required to be charged, and may transmit, to the server **200**, the determined required amount of charge. In this case, if the vehicle **100** receives power from the wireless charging station **320**, an operation of updating the required amount of charge based on the received power, and transmitting the updated required amount of charge to the server **200** may be further included.

[0062] The charging management device of the vehicle **100** may further include operations of calculating an estimated arrival time to the charge-available wireless charging station **320**, and transmitting the calculated estimated arrival time to the server **200**. The charging management device of the vehicle **100** may further include an operation of transmitting a location of the vehicle to the server **200** in real time.

[0063] The server **200** may be a charging management server for implementing a wireless charging service according to embodiments of the present disclosure. That is, the server **200** may be a server for managing wireless charging of a vehicle on a road.

[0064] The server **200** may receive various information and signals from the vehicle **100**. The server **200** may receive information on the traveling route or information on the current location and destination from the vehicle **100**. The server **200** may receive a charging execution request from the vehicle **100**. The charging execution request may be transmitted from the vehicle **100** immediately before traveling of the vehicle **100**. Alternatively, the charging execution request may be a signal generated based on the user of the vehicle **100** subscribing to the wireless charging service in advance.

[0065] The server **200** may identify at least one wireless charging station **320** through which the vehicle **100** passes on the traveling route of the vehicle **100** based on the information on the traveling route and the current location. In this case, the server **200** may confirm wireless charging stations **320** on the traveling route based on the information on the traveling route received from the vehicle **100** or based on the information on the traveling route calculated based on the information on the current location and destination received from the vehicle **100**.

[0066] The server **200** may transmit information on at least one identified wireless charging station **320** to the vehicle **100** and/or the corresponding charge control device **310** that controls the wireless charging station **320**. The server **200** may directly transmit information on the vehicle to the wireless charging station **320** instead of the charge control device **310**. For example, the server **200** may transmit a list of at least one wireless charging station **320** to the vehicle **100** and/or the charge control device **310**.

[0067] The server **200** may receive the amount of power charged in the battery of the vehicle **100** from the charge control device **310** in association with the information on the vehicle **100**. The server **200** may generate billing information for the vehicle based on the received amount of power. In this case, the billing information may be generated based on the sum of all amounts of power received by the vehicle **100** on the traveling route. The server **200** may transmit the billing information to a payment server of a credit card company or the like to charge the electricity bill corresponding to the amount of power used to charge the battery of the vehicle **100**. In order to charge the electricity bill, the server **200** may transmit, to the payment server, payment method information (e.g., credit card information) received from the user together with billing information.

[0068] The server **200** may communicate with the charging management device (or user device) of the vehicle **100** and the wireless charging system **300** through the network **500**.

[0069] The wireless charging system **300** is a system configured to enable wireless charging by wirelessly transmitting power to the battery of the vehicle **100**. The wireless charging system **300** may include the charge control device **310** and the wireless charging station **320**.

[0070] The wireless charging station **320** is a device installed under the ground of a road and configured to wirelessly transmit power within a predetermined region. The wireless charging station **320** may determine whether the vehicle **100** is stopped in a predetermined region. For example, the wireless charging station **320** may determine whether the vehicle **100** is stopped at a location in which a power transmission rate is equal to or greater than a reference value if power is wirelessly transmitted to the battery of the vehicle **100**. That is, determination may be made as to whether the wireless charging efficiency of the battery of the vehicle **100** by the wireless charging station **320** is equal to or greater than a reference value. If it is determined that the vehicle **100** is stopped in the predetermined region, the wireless charging station **320** may wirelessly transmit

power to the battery of the vehicle **100**.

[0071] The wireless charging station **320** may perform power transmission under the control of the charge control device **310**. In addition, the wireless charging station **320** may calculate the amount of power transmitted to the vehicle **100** and transmit the calculated amount of power to the charge control device **310**.

[0072] The charge control device **310** is a device configured to control the charging operation of the wireless charging station **320**. The charge control device **310** may be configured to control an operating state of the wireless charging station **320** based on vehicle information received from the server **200** and signal information of traffic lights. The vehicle information may include identification information for identifying the vehicle **100** and location information indicating the location of the vehicle **100**.

[0073] The charge control device **310** may be configured to communicate with the server **200** to receive the amount of power charged in the battery of the vehicle **100** from the wireless charging station **320**, and transmit the received amount of power to the server **200**. When transmitting the amount of power to the server **200**, the charge control device **310** may transmit it in association with vehicle information.

[0074] The charge control device **310** may be configured to control an operation state of the wireless charging station **320** to switch from a sleep state to a standby state when the signal information of the traffic lights indicates that the vehicle **100** is to be stopped in a traveling direction. In addition, the charge control device **310** may be configured to control the wireless charging station **320** to stop power transmission before a signal for the vehicle traveling direction in the traffic lights is changed from a stop signal to a go signal.

[0075] The wireless charging system **300** may communicate with the charging management device (or user device) of the vehicle **100** and the server **200** through the network **500**.

[0076] A signal controller **400** controls signals of traffic lights installed on roads such as intersections. The signal controller **400** may transmit signal information of the traffic lights to the wireless charging system **300**. The signal information may include a variety of information to determine signal states of the traffic lights, such as the current signal, the next signal, and the lengths of respective signals of the traffic lights.

[0077] In the present embodiment, the signal controller **400** has been illustrated as being directly connected to the wireless charging system **300** without going through the network **500**, but the present disclosure is not limited thereto. That is, the signal controller **400** may be connected to the wireless charging system **300** through the network **500**. Furthermore, although not illustrated, the signal controller **400** may be connected to a traffic control center or the like through the network **500** to control or monitor the operation.

[0078] The network **500** connects respective components in the system with each other to enable wireless and/or wired communication. The network **500** may be implemented by various methods in which respective components may be communicatively connected, and is not limited to a specific method.

[0079] FIG. **3** is a diagram illustrating a situation in which a system according to embodiments of the present disclosure operates. FIG. **3** illustrates in detail the situation of a specific intersection in the situation illustrated in FIG. **1**.

[0080] As described above, the system may include the vehicle **100**, the server **200**, and the wireless charging system **300**.

[0081] At the intersection, the wireless charging station **320** having a chargeable region **321** and the charge control device **310** for controlling the wireless charging station **320** may be provided. The chargeable region **321** may have a plurality of chargeable sub-regions **321a** to **321d**. In FIG. **3**, the chargeable region **321** is illustrated as having four chargeable sub-regions **321a** to **321d**; however, this is by example and is not limited thereto. The chargeable region **321** may have less than four chargeable sub-regions depending on road conditions, and may have five or more chargeable sub-

regions.

[0082] If the vehicle **100** is stopped to be completely covered by any one of the plurality of chargeable sub-regions **321a** to **321d**, the wireless charging efficiency will be higher than a reference value. Accordingly, the battery of the vehicle **100** will be charged through a wireless power transmission unit provided in the wireless charging station **320**. On the other hand, if the vehicle **100** is stopped over a part of any one of the plurality of chargeable sub-regions **321a** to **321d**, wireless charging efficiency is determined according to information such as a relative location between the wireless charging unit provided in the vehicle **100** and the wireless power transmitting unit of the chargeable sub-region. Then, if the determined wireless charging efficiency is equal to or greater than a reference value, the battery of the vehicle **100** is charged.

[0083] The server **200** may receive information such as traveling speed and current location information from the vehicle **100** in real time. In addition, the server **200** may further receive information on a battery state of charge (SOC), destination information, or the like, in addition to the information listed above. The above-described information received by the server **200** may be transmitted to the charge control device **310**. Alternatively, the above-described information may be transmitted directly from the vehicle **100** to the charge control device **310**.

[0084] The server **200** searches for a route to the destination of the vehicle **100** using the current location information and destination information received from the vehicle **100**. However, this is by example, and as described above, a traveling route may be searched for in a navigation system of the vehicle **100**, a navigation application of the user device, or the like. The charge control device **310** may receive the traveling route determined according to the searching result directly from the vehicle **100** or through the server **200**.

[0085] The server **200** identifies the wireless charging station **320** that is charge-available among the routes to the destination of the vehicle **100**. The server **200** transmits information on the vehicle **100**, such as the above-described current location information, traveling route, and traveling speed, to the corresponding charge control device **310** controlling the identified wireless charging station **320**.

[0086] The charge control device **310** of the wireless charging system **300** receiving information on the vehicle **100** from the server **200** receives signal information from the signal controller **400**. The signal information may include a variety of information to determine signal states of the traffic lights, such as the current signal, the next signal, the lengths of respective signals, and the period of the signals of the traffic lights. The signal information may be provided by the signal controller **400** in response to a request of the charge control device **310**. Alternatively, the signal information may be transmitted and stored periodically or in advance from the signal controller **400** to the charge control device **310**, irrespective of the request of the charge control device **310**. In this case, the charge control device **310** may be configured to receive an updated content only if the signal period or the like is changed (for example, if the signal is changed to a yellow blinking signal at night).

[0087] The charge control device **310** uses the distance from the current location of the vehicle **100** to the chargeable region **321** of the wireless charging station **320** and the traveling speed of the vehicle **100** to calculate the time it takes for the vehicle to arrive at the chargeable region **321**. The equation for calculating the time it takes for the vehicle **100** to arrive at the chargeable region **321** is as shown in Equation (1) below:

$$[00001] \quad T_{\text{arrival}(i)} = \frac{D_{i-1}}{V_{\text{current}}} \quad (1)$$

[0088] where D_{i-1} is the distance from the current location to the chargeable region.

[0089] Subsequently, the charge control device **310** uses a stop-go signal period in the chargeable region **321** and the time it takes to arrive at the chargeable region **321** to calculate a stop time when the vehicle **100** arrives at the corresponding chargeable region **321**. An equation for calculating the stop time when the vehicle **100** arrives at the corresponding chargeable region **321** is as shown in Equation (2) below:

[00002] $T_{\text{remain_stop_signal}} = T_{\text{current_stop_signal}} - [T_{\text{arrival}(i)} \bmod T_{(\text{Red, Green})}]$ (2) [0090] where $T_{(\text{Red, Green})}$ is the stop-go signal period, and [0091] $T_{\text{current_stop_signal}}$ is the time remaining until the next go signal [0092] ($\bmod$ is a modulo function that takes the remainder of division as a result). [0093] That is, the remaining stop time is calculated by subtracting the time when the vehicle has arrived at the current intersection from the time remaining until the go signal at the intersection. At this time, $T_{\text{sub.remain_stop_signal}} \leq T_{(\text{Red})}$ ($T_{(\text{Red})}$ is the length of a vehicle-stop signal). [0094] In applying Equation (2) above, the $T_{\text{sub.current_stop_signal}}$ value may be appropriately adjusted. For example, it is assumed that at a crossroad, the signal period of traffic lights is 2 minutes (that is, $T_{\text{sub.}(Red, Green)}$ is 2 minutes), a straight-ahead signal, which is the go signal, is 45 seconds, a left turn signal and a stop signal, which are vehicle-stop signals, are 15 seconds and 1 minute, respectively (the total length of the stop signal is 1 minute and 15 seconds). At this time, if the result value of $[T_{\text{sub.arrival}(i)} \bmod T_{\text{sub.}(Red, Green)}]$ is 10 seconds and $T_{\text{sub.current_stop_signal}}$ is 50 seconds (25 seconds elapsed after the signal of the traffic lights is turned to the stop signal), $T_{\text{sub.remain_stop_signal}}$ is 40 seconds. On the other hand, if the result value of $[T_{\text{sub.arrival}(i)} \bmod T_{\text{sub.}(Red, Green)}]$ is 40 seconds and $T_{\text{sub.current_stop_signal}}$ is 30 seconds (45 seconds elapsed after the signal of the traffic lights is turned to the stop signal), $T_{\text{sub.remain_stop_signal}}$ is -10 seconds. That is, the vehicle arrives at the crossroad after the stop signal has changed to the go signal, and thus charging is not possible. For another example, a case where the current signal of the traffic lights is the go signal may be considered. In this case, calculation is performed using the length of the current remaining go signal as the $T_{\text{sub.current_stop_signal}}$ value. If the result value is positive, charging is not possible since the go signal remains when the vehicle arrives at the crossroad. If the result value is negative, the signal is the stop signal when the vehicle arrives at the crossroad, and thus the value obtained by adding the result value to the total length of the stop signal may be set as $T_{\text{sub.remain_stop_signal}}$ (that is, the same as $T_{\text{sub.current_stop_signal}}$ set by adding the total length of the stop signal to the length of the remaining go signal in the calculation of Equation (2)). As described above, when performing the calculation of Equation (2), the value of $T_{\text{sub.current_stop_signal}}$ may be appropriately adjusted depending on the current signal type of the traffic lights, the remaining time of the signal, and the result value of $[T_{\text{sub.arrival}(i)} \bmod T_{\text{sub.}(Red, Green)}]$.

[0095] Subsequently, the charge control device **310** uses the stop time when the vehicle **100** arrives at the chargeable region **321** and the time when vehicles currently in the chargeable region **321** get out of the chargeable region **321** to calculate an estimated charging time of the vehicle **100** in the chargeable region **321**. The equation for calculating the estimated charging time of the vehicle **100** in the chargeable region **321** is as shown in Equation (3) below:

$$[00003] T_{\text{charge}(i)} = T_{\text{remain_stop_signal}} - T_{\text{leave}} \quad (3)$$

[0096] When being utilized, Equation (3) described above should be appropriately adjusted and changed depending on road conditions. This is because if a large number of vehicles are at a standstill at the intersection, the vehicle **100** of the user may be stopped outside the chargeable region **321** and may pass through the chargeable region **321** with other vehicles that have been stopped at the time of the go signal. On the other hand, there will be a case where another vehicle that has been stopped passes at the go signal and the vehicle **100** of the user fails to pass through the intersection at the corresponding signal and is stopped again in the chargeable region **321**. In this case, the above Equation (3) could be utilized.

[0097] Therefore, the charge control device **310** is to use Equation (3) in comprehensive consideration of the current road conditions, the average number of vehicles that have been stopped passing through the intersection at the time of the go signal, a pedestrian situation, or the like.

[0098] Subsequently, the server **200** receives the estimated charging time calculated from each charge control device **310**. Then, the server **200** calculates the total available charging time by summing all of the estimated charging times received from all charge control devices **310** on the

traveling route to the destination of the vehicle **100**. The equation for calculating the total available charging time is as shown in Equation (4) below:

$$[00004] \quad T_{\text{charge_total}} = \sum_{i=1}^N T_{\text{charge}(i)} \quad (4)$$

[0099] The server **200** may derive a charge amount that may be used for charging while the vehicle **100** is going to the destination based on the calculated total charging time, and transmit it to the vehicle **100**. In addition, the server **200** may transmit the total available charging time to the vehicle **100**. Alternatively, in addition to the amount of charge that is charge-available and the total available charging time, the server **200** may separately transmit the chargeable amount of charge and the estimated charging time for each wireless charging station **320** identified and provided as being charge-available to the vehicle **100** in traveling.

[0100] Information, such as the estimated charging time, the chargeable amount of charge, and the total available charging time, may be updated in real time by reflecting the current location information that is changed as the vehicle **100** travels. In addition, if the vehicle **100** performs charging in the chargeable region **321** of a specific wireless charging station **320** or passes the corresponding chargeable region **321**, by reflecting it, the information such as the estimated charging time, the chargeable amount of charge, and the total available charging time may be updated.

[0101] In the embodiment according to the present disclosure, it has been described that the server **200** serves as a medium for exchanging information between the vehicle **100** and the charge control device **310**, and the charge control device **310** calculates the estimated charging time and chargeable amount of charge; however, the present disclosure is not limited thereto. For example, the server **200** may receive corresponding signal information from the charge control devices **310** to directly calculate the estimated charging time and the chargeable amount of charge.

Alternatively, the vehicle **100** may directly exchange information with the charge control device **310** to perform all of the above-described functions, and the server **200** may be omitted.

[0102] Hereinafter, a detailed configuration and function of each component of a system according to the present disclosure will be described.

[0103] FIG. **4** is a block diagram illustrating a functional configuration of a charging management device according to embodiments of the present disclosure.

[0104] Referring to FIG. **4**, the vehicle **100** may include a vehicle controller **110**, a battery pack **120**, a wireless charging unit **130**, and a communication unit **140**.

[0105] The vehicle controller **110** controls the operation of various devices provided inside the vehicle **100**. The vehicle controller **110** may operate as a host controller of a battery management system (BMS) **122** provided in the battery pack **120**. The vehicle controller **110** may be communicatively connected with the BMS **122** to control the charging and discharging of the battery module **121** by controlling the BMS **122** and monitor the state of the battery module **121**.

[0106] The battery pack **120** may be made of one or more battery cells, and may include the battery module **121** which is capable of performing charging and discharging, and the BMS **122** that performs control such that overcharging, overdischarging, and the like are prevented by monitoring the voltage, current, temperature, or the like of the battery cell and/or the battery module **121**. In addition, although not illustrated, the battery pack **120** may further include a switching circuit for controlling the charge/discharge current flow of the battery module **121** and a battery protection unit (BPU) for protecting the battery pack **120** when an abnormality occurs in the battery pack **120**.

[0107] The battery module **121** includes one or more battery cells capable of charging and discharging. In the battery module **121**, a plurality of battery cells may be connected in series and/or in parallel to each other depending on the required specifications of the battery pack **120**. That is, the number and connection type of battery cells may be determined depending on the required output (voltage, current, or the like) of the battery pack **120**. The battery cell may be a lithium ion (Li-ion) battery, a lithium ion polymer battery, a nickel cadmium (Ni—Cd) battery, a

nickel hydrogen (Ni-MH) battery, or the like, and is not limited thereto as long as the battery is a rechargeable battery.

[0108] The BMS **122** controls and manages the overall operation of the battery pack **120**. The BMS **122** may control the operation of the switching circuit in order to control the charging/discharging operation of the battery module **121**. In addition, the BMS **122** may monitor the voltage, current, temperature, and the like of each battery cell included in the battery module **121** and/or the battery module **121**. In addition, for monitoring by the BMS **122**, a sensor or various measurement modules, which are not illustrated, may be additionally installed in any location such as the battery module **121**, the charge/discharge path, or the battery pack **120**. The BMS **122** may calculate a parameter indicating the state of the battery module **121**, for example, SOC or SOH, based on measurement values such as the monitored voltage, current, and temperature.

[0109] The battery pack **120** may be communicatively connected with the vehicle controller **110**, which is a host controller. That is, the BMS **122** may transmit various data on the battery pack **120** to the vehicle controller **110**, and may receive a control signal regarding the operation of the battery pack **120** from the vehicle controller **110**.

[0110] The communication unit **140** is a device configured to communicate with an external device. The communication unit **140** may exchange information by communicating with the server **200** functioning as a charging management server through the network **500**. For example, the communication unit **140** may transmit, to the server **200**, information on the battery module **121** transmitted from the BMS **122** to the vehicle controller **110**, for example, information such as SOC. In addition, the communication unit **140** may transmit, to the server **200**, information such as the charging execution request generated based on the input or setting of the user, a current location, the traveling route, and the traveling speed of the vehicle **100**.

[0111] The wireless charging unit **130** is a power receiving device configured to receive power wirelessly transmitted from the outside. The wireless charging unit **130** receives power from a power transmission unit of the wireless charging station **320** if the vehicle **100** is stopped in the chargeable region **321** provided in the wireless charging station **320** of the wireless charging system **300**. The wireless charging unit **130** may charge the battery module **121** using the received power.

[0112] FIG. 5 is a block diagram illustrating a hardware configuration of the charging management device according to embodiments of the present disclosure.

[0113] The charging management device **101** of the vehicle **100** including the vehicle controller **110**, the battery pack **120**, the wireless charging unit **130**, and the communication unit **140** may include at least one processor, at least one memory, a communication device, and a power receiving device in terms of hardware.

[0114] Referring to FIG. 5, the charging management device **101** may include a processor (CPU) **150**, a memory **151**, a communication interface **152**, an input/output interface **153**, and the like.

[0115] The processor **150** is the at least one processor described above, and may be any of various controllers capable of performing an operation by executing computer program instructions such as a CPU, MCU, microcontroller, or microprocessor. The processor **150** may process various operations and arithmetic operations in the charging management device **101** and may control each component.

[0116] The memory **151** may store an operating system program and a computer program for performing functions of the vehicle controller **110**. The memory **151** may include a volatile memory and a nonvolatile memory. For example, the memory **151** may be at least one of various storage media such as a semiconductor memory, for example, RAM, ROM, and flash memory, a magnetic disk, and an optical disk. The memory **151** may be a memory built into the processor **150** or may be an additional memory installed separately from the processor **150**.

[0117] The communication interface **152** is configured to communicate with the outside by wire and/or wirelessly.

[0118] The input/output interface **153** performs input/output of various input signals and output signals. For example, the processor **150** included in the charging management device **101** may receive signals from various sensors through the input/output interface **153**. The input/output interface **153** may further include an input device configured to receive an input from the user, and an output device such as a display for displaying various signals or information generated by the charging management device **101**. In addition, in the charging management device **101** according to the present disclosure, a device for a wireless charging circuit through which power is received may be included as the input/output interface **153**.

[0119] The processor **150** may implement a module that performs various functions of the charging management device **101** by executing a program stored in the memory **151**. The processor **150** may operate together with the communication interface **152** to function as the communication unit **140**. In addition, the processor **150** may operate together with the input/output interface **153** to function as the wireless charging unit **130**.

[0120] Specifically, the processor **150** as at least one processor executes computer program instructions stored in the memory **151** to perform operations for charging management.

Specifically, the traveling route of the vehicle **100** to the destination is searched for based on the destination information input by the user and the current information of the vehicle **100**. There may be a plurality of traveling routes to be searched for. For example, there may be a plurality of traveling routes to be searched for according to conditions such as the optimum road, a free road, a highway priority, the shortest distance, and the minimum time. The searching for the traveling route may be performed by a navigation system built in the vehicle **100** or a navigation application installed in a user device. One of the plurality of traveling routes searched may be determined as a traveling route on which the vehicle **100** travels.

[0121] If the processor **150** transmits information on the traveling route to the server **200**, the server **200** identifies wireless charging stations **320** that are charge-available to the vehicle **100** while being at a standstill from the server **200**. Alternatively, the processor **150** may transmit information on the current location and destination of the vehicle **100** to the server **200**, the server **200** directly searches for the traveling route, and then the vehicle **100** may identify wireless charging stations **320** that are charge-available.

[0122] If the wireless charging stations **320** that are charge-available to the vehicle **100** while being at a standstill are identified, the corresponding wireless charging stations **320** are provided to the charging management device **101**. Accordingly, when the vehicle **100** is stopped at intersections in which the identified wireless charging stations **320** are installed and is stopped at chargeable locations, the vehicle **100** receives power from the wireless charging stations **320**.

[0123] The processor **150** receives, from BMS **122**, SOC information or information used to calculate SOC. Further, the processor **150** determines a required amount of charge required for charging based on the received SOC or the calculated SOC. The determined required amount of charge may be transmitted to the server **200**.

[0124] If charging is performed by receiving power from the wireless charging station **320**, the processor **150** may update a required amount of charge based on the received power. In addition, the updated required amount of charge may be transmitted back to the server **200**.

[0125] In addition, the processor **150** may calculate an estimated arrival time to each of the wireless charging stations **320** identified based on the current location, the traveling route, and the traveling speed of the vehicle **100**. The processor **150** may transmit the calculated estimated arrival time to the server **200**. The required amount of charge and the estimated arrival time transmitted to the server **200** may be transmitted to the charge control device **310** that controls each wireless charging station **320** as necessary.

[0126] On the other hand, although not illustrated, the charging management device **101** may include a global positioning system (GPS) device in order to determine the current location of the vehicle **100**. The current location of the vehicle **100** determined by the GPS device may be

transmitted from the charging management device **101** to the server **200** in real time or periodically. In addition, the charging management device **101** may determine the current location of the vehicle **100** in real time and may update the traveling route in real time or periodically based on the current location of the vehicle **100**.

[0127] FIG. **6** is a flowchart illustrating an operation of the charging management device according to embodiments of the present disclosure.

[0128] Referring to FIG. **6**, the user may apply for a charging service capable of performing charging using the wireless charging system **300** on the road while the vehicle **100** is traveling (S**100**). The application for the charging service may be made immediately before traveling of the vehicle **100** by the user. Alternatively, the user may apply in advance to receive the charging service for the vehicle **100**. In this case, the charging service may be automatically provided whenever traveling of the vehicle **100** is started. The application for the charging service may be made to the server **200**.

[0129] The charging management device **101** searches for a traveling route if the user tries to move to the destination using the vehicle **100** (S**102**). The charging management device **101** may search for the traveling route based on the current location and destination of the vehicle **100**. In addition, the charging management device **101** may directly search for the traveling route, but may also search for the traveling route in the navigation application of the user terminal or the server **200**.

[0130] The traveling route determined based on the plurality of traveling routes searched is transmitted to the server **200** (S**104**). Of course, as described above, if the server **200** searches for the traveling route, the charging management device **101** may transmit, to the server **200**, information on the current location and destination of the vehicle **100**.

[0131] If the server **200** identifies the wireless charging stations **320** that are charge-available while being at a standstill on the traveling route based on the traveling route, the charging management device **101** receives a list of the identified wireless charging stations **320** (S**106**).

[0132] The vehicle **100** may travel toward the destination along the traveling route, and traveling information collected in this process may be transmitted to the server **200** by the charging management device **101** (S**108**). The traveling information may include information such as the current location and the traveling speed of the vehicle **100**. In addition, the charging management device **101** may further transmit, along with the traveling information, information on a required amount of charge and the like calculated based on the SOC of the battery module **121**, the traveling distance to the destination, or the like.

[0133] If the vehicle **100** is stopped by a stop signal at an intersection in traveling, determination is made as to whether the vehicle **100** has stopped in the chargeable region (S**110**). This determination may be performed by the charging management device **101** or may be performed by the wireless charging system **300**. In addition, if the vehicle is stopped in the chargeable region and the wireless charging efficiency from the wireless charging station **320** to the battery module **121** is equal to or greater than a reference value, charging is performed (S**112**). Additionally, when the charging execution request is transmitted from the charging management device **101** of the vehicle **100** to the wireless charging system **300** before starting charging, charging may actually be started. That is, charging may be performed only when the user finally decides whether to perform charging, and the charging execution request requesting that charging be performed is transmitted from the charging management device **101** to the wireless charging system **300**.

[0134] When charging the vehicle **100**, the charging management device **101** may transmit identification information allowing the vehicle **100** to be identified, to the wireless charging station **320** that transmits power or the charge control device **310** that controls it. Alternatively, the charge control device **310** may include a device capable of recognizing the vehicle **100** such as a camera to identify the vehicle **100**. Since the charge control device **310** secures identification information allowing the vehicle **100** to be identified, it will be possible to impose the electricity bill for power used for charging on the vehicle **100** in the future.

[0135] Steps S108 to S116 are repeated until the vehicle **100** arrives at the destination (Yes in S114) or until it is determined that charging of the battery module **121** is no longer required (No in S116). [0136] In this way, it is possible to wirelessly charge the battery even if the vehicle **100** is temporarily stopped in traveling to the destination, thereby enabling efficient charging scheduling. [0137] FIG. 7 is a block diagram illustrating a functional configuration of the server according to embodiments of the present disclosure.

[0138] The server **200** is a server configured to provide a wireless charging service. In particular, the server **200** is a server for managing wireless charging of a vehicle on a road. Referring to FIG. 7, the server **200** may include a control unit **210**, a communication unit **220**, a billing unit **230**, a charging management unit **240**, and a traffic information management unit **250**.

[0139] The control unit **210** controls operations of the communication unit **220**, the billing unit **230**, the charging management unit **240**, and the traffic information management unit **250**, which are components of the server **200**. When the control unit **210** receives a traveling route from the charging management device **101**, the control unit **210** identifies the wireless charging stations **320** that are charge-available to the vehicle **100** on the traveling route based on the received traveling route. If the control unit **210** receives information on the current location and destination instead of the traveling route from the charging device **101**, the control unit **210** directly searches for the traveling route. Then, based on the traveling route searched, the wireless charging stations **320** are identified.

[0140] The communication unit **220** is a device configured to communicate with external devices based on the control of the control unit **210**. The communication unit **220** may exchange information by communicating with the charging management device **101** of the vehicle **100** through the network **500**. For example, the communication unit **220** may receive, from the charging management device **101**, SOC information of the battery module **121**, information on the required amount of charge, information and/or signals on the current location, the traveling route, the traveling speed, the charging execution request, or the like.

[0141] The communication unit **220** may also communicate with the wireless charging systems **300** through the network **500**. The communication unit **220** may transmit the list of wireless charging stations **320** identified by the control unit **210** to the charging management device **101**. In addition, the communication unit **220** may transmit information on the vehicle **100** to which the wireless charging service is provided, to the charge control device **310** that controls each of the identified wireless charging stations **320**. For example, the communication unit **220** may transmit identification information for identifying the vehicle **100**, the current location, the traveling speed, the traveling route, the required amount of charge of the vehicle **100**, and the estimated arrival time it takes the vehicle **100** to arrive at the corresponding wireless charging station **320**. Of course, instead of the information listed above being transmitted to the charge control device **310** by the communication unit **220**, a part or all of the information may be directly transmitted to the charge control device **310** from the charging management device **101** of the vehicle **100**.

[0142] The communication unit **220** may receive the estimated charging time and the available amount of charge of the vehicle **100** from the charge control device **310**. The information as mentioned above may be transmitted before the vehicle **100** arrives at the wireless charging station **320** managed by the charge control device **310** or may be transmitted during charging. In addition, the communication unit **220** may receive the amount of power actually charged to the vehicle **100** from the charge control device **310**. The amount of power may be a value corresponding to the amount of power transmitted to the vehicle **100** by the wireless charging station **320**.

[0143] The billing unit **230** is configured to generate billing information based on information on the received amount of power. The billing unit **230** may generate billing information by associating identification information for identifying the vehicle **100** or the vehicle **100** transmitted from the user, payment method information, and the like, with information on the amount of power. The generated billing information may be transmitted to a payment server of a credit card company or

the like to charge the electricity bill corresponding to the amount of power used to charge the battery module **121** of the vehicle **100**.

[0144] The charging management unit **240** may receive, from a plurality of wireless charging systems **300**, information on the estimated charging time and the charge-available amount of power of the vehicle **100** at each wireless charging station **320**. In addition, the charging management unit **240** may calculate a total available charging time, a total chargeable amount of power, and the like based on the received information. Further, the charging management unit **240** may perform scheduling for charging of the vehicle **100** in traveling based on the received information, calculated information, or the like. At this time, the charging management unit **240** may receive information on traffic conditions from the traffic information management unit **250** and may perform scheduling for charging the vehicle **100** by reflecting the received information on traffic conditions.

[0145] The traffic information management unit **250** collects information on traffic conditions of the road. The information on traffic conditions may include information on an accident or construction, information on weather that affects the traffic flow, or the like, as well as the amount of movement of the vehicle. The traffic information management unit **250** may collect related information from an external agency (meteorological agency, CCTV of a road, government offices that manage the road, or the like).

[0146] FIG. **8** is a block diagram illustrating a hardware configuration of the server according to embodiments of the present disclosure.

[0147] The server **200** including the control unit **210**, the communication unit **220**, the billing unit **230**, the charging management unit **240**, the traffic information management unit **250**, and the like, may include at least one processor, at least one memory, a communication device, and the like in terms of hardware.

[0148] Referring to FIG. **8**, the server **200** may include a processor (CPU) **260**, a memory **261**, a communication interface **262**, an input/output interface **263**, and the like.

[0149] The processor **260** is the at least one processor described above, and may be any of various controllers capable of performing an operation by executing computer program instructions such as a CPU, MCU, microcontroller, or microprocessor. The processor **260** may process various operations and arithmetic operations in the server **200** and may control each component.

[0150] The memory **261** may store the operating system program and a computer program for performing the functions of the control unit **210**, the communication unit **220**, the billing unit **230**, the charging management unit **240**, and the traffic information management unit **250**. The memory **261** may include a volatile memory and a nonvolatile memory. For example, the memory **261** may be at least one of various storage media such as a semiconductor memory, for example, RAM, ROM, and flash memory, a magnetic disk, and an optical disk. The memory **261** may be a memory built into the processor **260** or may be an additional memory installed separately from the processor **260**.

[0151] The communication interface **262** is configured to communicate with the outside by wire and/or wirelessly.

[0152] The input/output interface **263** performs input/output of various input signals and output signals.

[0153] The processor **260** may execute the program stored in the memory **261** to implement the module that performs the functions of the control unit **210**, the billing unit **230**, the charging management unit **240**, and the traffic information management unit **250** of the server **200**. The processor **260** may operate together with the communication interface **262** to function as the communication unit **220**.

[0154] Specifically, the processor **260** as at least one processor executes computer program instructions stored in the memory **261** to perform operations for providing the wireless charging service. Specifically, the server **200** receives information on at least the traveling route and the

current location of the vehicle from the vehicle **100** through the communication unit **220**.

Alternatively, instead of receiving the traveling route, the server may receive information on the current location and destination, and may directly search for the traveling route based on the received information. The server **200** may identify at least one wireless charging station **320** through which the vehicle **100** passes on the traveling route based on the received information on the traveling route and the current location of the vehicle. In addition, the server **200** provides information on the at least one identified wireless charging station **320** to the charging management device **101** of the vehicle **100**.

[0155] The server **200** transmits the information on the vehicle **100** to a corresponding vehicle control device **310** that controls the at least one identified wireless charging station **320**. The information on the vehicle **100** may include identification information for identifying the vehicle **100**, the current location, the traveling route, the traveling speed, and the estimated arrival time of the vehicle **100**, and the like.

[0156] In addition, the server **200** receives the amount of power charged in the battery module **121** of the vehicle **100** from the charge control device **310** in association with the identification information of the vehicle **100**. Then, the server **200** may generate billing information for the vehicle **100** based on the received amount of power. An operation of generating the billing information may be generated based on the sum of all amounts of power received by the vehicle **100** on the traveling route.

[0157] FIG. **9** is a flowchart illustrating an operation of the server according to embodiments of the present disclosure.

[0158] The server **200** receives the application for the charging service for the vehicle **100** from the user (**S200**). As described above, the application for the charging service may be made immediately before traveling of the vehicle **100** by the user. Alternatively, the user may apply in advance to receive the charging service for the vehicle **100**. In this case, the charging service may be automatically provided whenever traveling of the vehicle **100** is started.

[0159] If the vehicle **100** is a vehicle which has applied for the charging service, the server **200** receives the traveling route from the vehicle **100** (**S202**). Alternatively, the server **200** directly searches for the traveling route by receiving information on the current location and destination of the vehicle **100**.

[0160] The server **200** identifies at least one wireless charging station **320** that is charge-available to the vehicle **100** while being at a standstill on the traveling route based on the traveling route (**S204**). Information on the identified wireless charging station **320** is transmitted to the vehicle **100** (**S206**).

[0161] If the vehicle **100** starts traveling, the server **200** receives various traveling information from the vehicle **100** (**S208**). The traveling information received by the server **200** may include the current location and the traveling speed. In addition, the server **200** may receive, from the vehicle, information and/or signals such as the required amount of charge and SOC information.

[0162] The server **200** transmits a part or all of the received vehicle information to the charge control device **310** (**S210**). Then, the server **200** may receive, from the charge control device **310**, the estimated arrival time at which the vehicle **100** is estimated to arrive at the wireless charging station **320**, the chargeable amount of charge calculated based on the estimated arrival time, and the like, in real time or periodically while the vehicle **100** is traveling.

[0163] If the vehicle **100** starts from a specific wireless charging station **320** and gets out of the chargeable region, the server **200** receives the amount of charging power from the charge control device **310** (**S212**). Then, the server **200** generates billing information based on the received amount of charging power (**S214**). The billing information may include the amount of charging power, vehicle identification information, payment method information, and the like. If charging is completed in all the wireless charging stations **320** when the vehicle **100** arrives at the destination and the server **200** receives the amounts of charging power from the corresponding charge control

devices **310**, the server **200** may add all the received amounts of charging power to calculate the final amount of charging power to be charged.

[0164] The server **200** generates billing information based on the amount of power, and transmits it to the payment server to charge the electricity bill (S216).

[0165] In this way, it is possible to wirelessly charge the battery even if the vehicle **100** is temporarily stopped in traveling to the destination, thereby enabling efficient charging scheduling. In addition, the server **200** may create profit by charging vehicles traveling on the road.

[0166] FIG. **10** is a block diagram illustrating a functional configuration of the wireless charging system according to embodiments of the present disclosure.

[0167] Referring to FIG. **10**, the wireless charging system **300** includes the charge control device **310** and the wireless charging station **320**.

[0168] The charge control device **310** may include a control unit **311** and a communication unit **312**.

[0169] The communication unit **312** is a device configured to communicate with the server **200** and the wireless charging station **320** under the control of the control unit **311**. In addition, through the communication unit **312**, the charge control device **310** may receive signal information of the traffic lights **410** from the signal controller **400**.

[0170] The control unit **311** is configured to control the operation of the wireless charging station **320**. The control unit **311** is configured to control an operating state of the wireless charging station **320** based on vehicle information received from the server **200** and signal information of the traffic lights. The control unit **311** may receive, from the server **200**, identification information for identifying the vehicle **100**, location information indicating the current location of the vehicle, and the like, as the vehicle information.

[0171] The control unit **311** may receive, from the server **200**, information such as the current location, the traveling route, and the traveling speed for the vehicle **100** that has applied for the charging service. The control unit **311** calculates the estimated arrival time at which the vehicle **100** arrives at the chargeable region **321** of the wireless charging station **320** based on the received information such as the current location, the traveling route, and the traveling speed. Then, the control unit **311** calculates an estimated charging time available to the vehicle **100** for charging in the same manner as described in FIG. **3**.

[0172] The control unit **311** may receive the amount of power charged in the vehicle **100** from the wireless charging station **320**. The control unit **311** may transmit, to the server **200**, the received amount of power and vehicle information in association with each other. In this case, the control unit **311** may identify the vehicle **100** that has been stopped in the chargeable region **321** in order to confirm whether the vehicle **100** currently stopped in the chargeable region **321** of the wireless charging station **320** is the vehicle **100** transmitted from the server **200**. For example, the charge control device **310** may identify the vehicle **100** by recognizing the license plate of the vehicle **100** through a camera or the like. Alternatively, the charge control device **310** may communicate with the charging management device **101** of the vehicle **100** or may communicate with a user device to identify the vehicle **100**. Still alternatively, the wireless charging station **320** may communicate with the vehicle **100** to identify the vehicle **100**, and the charge control device **310** may receive the result. If the vehicle **100** identified by the wireless charging system **300** matches the vehicle information transmitted from the server **200**, the vehicle **100** may be charged.

[0173] The control unit **311** may be configured to control an operation state of the wireless charging station **320** such that the operation state thereof is switched from a sleep state to a standby state when the signal information of the traffic lights **410** indicates that the vehicle **100** is to be stopped in a traveling direction. In addition, the control unit **311** may be configured to control the wireless charging station **320** to stop power transmission before a signal for the vehicle traveling direction in the traffic lights **410** is changed from a stop signal to a go signal. Through this control, power consumption of the wireless charging station **320** may be minimized. In addition, power

transmission may be stopped before the vehicle **100** starts, thereby preventing unnecessary waste of power.

[0174] The wireless charging station **320** is a facility on the road which is configured to wirelessly transmit power to the vehicle **100**. The wireless charging station **320** may be installed under the ground of the road. A wireless power transmitting unit capable of wirelessly transmitting power is provided in a predetermined region under the ground of the road. When the vehicle **100** is stopped within a predetermined region, power may be wirelessly transmitted to the vehicle **100**.

[0175] The wireless charging station **320** may include a control unit **322**, a communication unit **323**, and a power transmission unit **324**.

[0176] The control unit **322** controls the operation of each component of the wireless charging station **320**. The control unit **322** determines the location of the vehicle **100** when the vehicle **100** enters the intersection and stops in the chargeable region **321**. If power is wirelessly transmitted to the vehicle **100**, the control unit **322** determines whether the vehicle **100** is stopped at a location in which the power transmission rate is equal to or greater than the reference value. That is, the control unit **322** functions as a vehicle location determination device. The power transmission rate equal to or greater than the reference value may mean that the power transmission rate when power is wirelessly transmitted between the power transmission unit **324** and the wireless charging unit **130** is equal to or greater than the reference value. The power transmission rate may mean wireless charging efficiency indicating power charged in the battery module **121** of the vehicle **100**, in the power transmitted from the power transmission unit **324**.

[0177] The communication unit **323** is a device configured to exchange information and control signals with the vehicle **100**, the charge control device **310**, and the like.

[0178] The power transmission unit **324** corresponds to a wireless power transmission device that wirelessly transmits power to the vehicle **100**. The power transmission method between the power transmission unit **324** and the wireless charging unit **130** may be implemented by various known methods, and is not limited to a specific method.

[0179] Meanwhile, if the power transmission unit **324** transmits power to the vehicle **100**, the control unit **322** calculates the amount of power transmitted. That is, the control unit **322** functions as a power amount calculation unit.

[0180] The charging management device **101** may receive, from the wireless charging station **320**, the amount of charging power, which is the amount of power which the wireless charging station **320** has transmitted to the vehicle **100**. The charging management device **101** may provide the received amount of charging power to the server **200** together with the identification information of the vehicle **100** to generate the billing information.

[0181] FIG. **11** is a flowchart illustrating an operation of the wireless charging system according to embodiments of the present disclosure. FIG. **12** is a flowchart illustrating a method for calculating an available charging time of a vehicle by the wireless charging system according to embodiments of the present disclosure.

[0182] First, referring to FIG. **11**, the charge control device **310** receives, from the server **200**, information on the vehicle **100** that has applied for the charging service (S300). In addition, the charge control device **310** receives signal information from the signal controller **400** (S302). Based on the received information on the vehicle **100** and signal information, the charge control device **310** calculates a chargeable time of the vehicle **100** at the corresponding wireless charging station **320** (S304).

[0183] Here, referring to FIG. **12**, a method for calculating the available charging time of the vehicle **100** is illustrated in detail. The charge control device **310** receives information such as the traveling speed and the current location from the server **200** (or alternatively the vehicle **100**) (S400). Then, the time it takes for the vehicle **100** to arrive at the chargeable region is calculated (S402). Then, a stop time to be stopped when the vehicle **100** arrives at the chargeable region is calculated (S404). Based on the calculated stop time, the available charging time, that is, the

estimated charging time is calculated (S406). The available charging time, which is the calculated estimated charging time, is transmitted to the server **200** or the vehicle **100** (S408). The method for calculating the estimated charging time of FIG. **12** has been described in detail with reference to FIG. **3**, and thus further description will be omitted.

[0184] Referring back to FIG. **11**, if the time when the vehicle **100** arrives at the chargeable region **321** is calculated, the charge control device **310** performs control such that the wireless charging station **320** switches from a sleep state to a standby state (S306). Then, determination is made as to whether the vehicle **100** is stopped at the wireless charging station **320** (S308). If the stopped vehicle is the vehicle that has applied for the charging service, determination is made as to whether the vehicle is stopped at the chargeable location (S310). That is, determination is made as to whether the power transmission rate is equal to or greater than the reference value.

[0185] If it is determined that the vehicle **100** is stopped at a chargeable location, charging is performed and power is wirelessly transmitted to the vehicle **100** (S312). The wireless charging station **320** calculates the amount of power transmitted to the vehicle **100** (S314), and transmits the amount of power to the server **200** for billing (S316).

[0186] In this way, it is possible to wirelessly charge the battery even if the vehicle **100** is temporarily stopped in traveling to the destination, thereby enabling efficient charging scheduling. In addition, the wireless charging system **300** may create profit by wirelessly transmitting power to vehicles that are stopped on the road in traveling.

[0187] That is, with the server **200** and the wireless charging system **300** according to embodiments of the present disclosure, it is possible to provide a wireless charging service that wirelessly charges the vehicle **100** while the vehicle **100** is at a standstill by using the plurality of wireless charging stations **320** provided under the ground of the road. The wireless charging service may include confirming, at the server **200**, at least one wireless charging station available to make charging, among the plurality of wireless charging stations, based on a traveling route of the vehicle, transmitting power from the wireless charging station to the vehicle if the vehicle is stopped in a charging available position at an intersection in which the at least one wireless charging station is installed, and generating billing information based on an amount of power transmitted to the vehicle.

[0188] In the method for providing the service, the method may further include receiving payment method information from the vehicle, and transmitting payment information to a payment server based on the payment method information and the billing information. Here, the billing information may be generated based on vehicle information recognized by identifying a license plate of the vehicle.

[0189] In embodiments of the present disclosure, the wireless charging station **320** has been described as communicating with the server **200** through the charge control device **310**; however, the present disclosure is not limited thereto. For example, the communication unit **323** of the wireless charging station **320** may directly communicate with the server **200** to receive at least some of the above-described information, and may transmit it to the charge control device **310**.

[0190] Further, in FIG. **3**, a configuration in which one charge control device **310** is connected to one wireless charging station **320** has been illustrated; however, the configuration is not limited thereto. For example, a configuration may be implemented in which a plurality of wireless charging stations **320** are installed at an intersection and one charge control device **310** controls the plurality of wireless charging stations **320**.

[0191] FIG. **13** is a block diagram showing another example of a functional configuration of the wireless charging system according to embodiments of the present disclosure.

[0192] As an alternative embodiment, the wireless charging system **300** may include a first communication unit **330** that receives, from the vehicle **100**, battery SOC information, a traveling speed, current location information, and destination information, current location information, and destination information, a route calculation unit **334** that calculates a route to the destination of the

vehicle **100** using the current location information and the destination information, and a charging time calculation unit **336** that calculates the available charging time of the vehicle **100** using information on chargeable regions while waiting for a signal located on the route to the destination. [0193] The first communication unit **330** further receives battery SOC information, traveling speeds, current location information, and destination information from a plurality of vehicles different from the vehicle **100**. The charging time calculation unit **336** may calculate an available charging time of the vehicle **100** by further considering traveling speeds and current location information of a plurality of other vehicles.

[0194] The charging time calculation unit **336** calculates at least one chargeable region through which the vehicle **100** passes by using the route to the destination of the vehicle **100**. Further, the charging time calculation unit **336** uses the distance from the current location of the vehicle **100** to the chargeable region and the traveling speed of the vehicle **100** to calculate the time it takes for the vehicle **100** to arrive at the chargeable region.

[0195] The charging time calculation unit **336** calculates the stop time when the vehicle **100** arrives at the chargeable region by using the stop-go signal period in the chargeable region and the time taken until the available charging time. In addition, the charging time calculation unit **336** may calculate the estimated charging time in the chargeable region of the vehicle **100** by using the stop time when the vehicle **100** arrives at the chargeable region and the current number of vehicles in the corresponding chargeable region.

[0196] The charging time calculation unit **336** calculates estimated charging times for all chargeable regions on the route to the destination of the vehicle **100**. In addition, the charging time calculation unit **336** calculates the total charging time available while the vehicle **100** is traveling to the destination by adding all the estimated charging times in respective chargeable regions. The first communication unit **330** may transmit, to the vehicle **100**, the calculated estimated charging time, estimated amount of charge, or total charging time for the vehicle.

[0197] The wireless charging system **300** may further include a second communication unit **338** for receiving a stop and go signal period in the chargeable region from the signal controller.

[0198] On the other hand, traffic conditions on the road may change from time to time. Due to factors such as accidents, construction, traffic jams, and an increase in the number of pedestrians, or the like, the available charging time calculated while the vehicle is traveling will change in real time. Accordingly, the charging time calculation unit **336** repeatedly calculates and updates the above charging time and the total available charging time based thereon in a predetermined period.

[0199] A storage unit **332** stores received SOC information, traveling speed, current location information, and destination information of each of the plurality of vehicles. In addition, the storage unit **302** stores equation information used to calculate the charging time of the vehicle. In addition, the storage unit **302** receives and stores signal change period information from a signal management device **50** located in a charging region **40** through the second communication unit **308**.

[0200] Terms such as “include”, “comprise”, or “have” described above mean that the corresponding component can be present unless otherwise stated, and thus it should be construed that other components may be further included rather than excluding other components. All terms including technical or scientific terms may be construed as having the same meaning as commonly understood by those of ordinary skill in the art to which the embodiments of the present disclosure belong, unless otherwise defined. Terms commonly used such as those defined in dictionaries should be interpreted as having a meaning that is consistent with their meaning in the context of the related art and will not be interpreted in an idealized or overly formal sense unless expressly so defined in the present disclosure.

[0201] The above description is merely illustrative of the technical idea of the present invention, and those of ordinary skill in the art to which the present invention pertains will be able to make various modifications and variations without departing from the essential characteristics of the

present invention. Accordingly, the embodiments disclosed in the present invention are not intended to limit the technical idea of the present invention, but to explain the technical idea, and the scope of the technical idea of the present invention is not limited by these embodiments. The scope of protection of the present invention should be interpreted by the accompanying claims, and all technical ideas within the scope equivalent thereto should be construed as being included in the scope of the present disclosure.

Claims

1. A wireless charging system comprising: a wireless charging station installed under the ground on a road and configured to wirelessly transmit power within a predetermined region of the road; and a charge control device configured to control charging of the wireless charging station, wherein the charge control device includes: a communication unit configured to communicate with a server to receive whether a power transmission rate for a location of a vehicle when the vehicle is stopped at a chargeable position of the wireless charging station is equal to or greater than a reference value; and a control unit configured to control an operation state of the wireless charging station based on vehicle information and whether the power transmission rate is equal to or greater than the reference value received from the server.
2. The wireless charging system of claim 1, wherein the vehicle information includes identification information for identifying the vehicle and location information indicating the location of the vehicle.
3. The wireless charging system of claim 1, wherein the wireless charging station includes: a vehicle location determination device for determining whether the vehicle is stopped within the predetermined region of the road, wherein the predetermined region of the road includes the chargeable position of the wireless charging station; a wireless power transmission unit configured to wirelessly transmit power to the vehicle stopped within the predetermined region of the road; and a power amount calculation unit configured to calculate an amount of power transmitted by the wireless power transmission unit to the vehicle.
4. The wireless charging system of claim 3, wherein the charge control device is configured to receive, from the wireless charging station, the amount of power charged in the vehicle, and transmit, to the server, the received amount of power in association with the vehicle information.
5. The wireless charging system of claim 1, wherein the charge control device is configured to control the operation state of the wireless charging station to switch from a sleep state to a standby state when a signal information of traffic lights on the road indicates that the vehicle is to be stopped due to the traffic lights in a traveling direction.
6. The wireless charging system of claim 1, wherein the charge control device is configured to control the wireless charging station to stop power transmission before a signal for a traveling direction of the vehicle in traffic lights on the road is changed from a stop signal to a go signal.
7. An operating method of a wireless charging system, comprising: communicating, by a communication unit, with a server to receive whether a power transmission rate for a location of a vehicle when the vehicle is stopped at a chargeable position of a wireless charging station of the wireless charging system is equal to or greater than a reference value; controlling, by a control unit of the wireless charging system, an operation state of the wireless charging station based on vehicle information and whether the power transmission rate is equal to or greater than the reference value received from the server; and wirelessly transmitting, by a wireless charging station of the wireless charging system installed under the ground on a road, power within a predetermined region of the road based on the operation state.
8. The operating method of claim 7, wherein the vehicle information includes identification information for identifying the vehicle and location information indicating the location of the vehicle.

9. The operating method of claim 7, further comprising: determining whether the vehicle is stopped within the predetermined region of the road, wherein the predetermined region of the road includes the chargeable position of the wireless charging station; wirelessly transmitting power to the vehicle stopped within the predetermined region of the road; and calculating an amount of power transmitted by the wireless power transmission unit to the vehicle.

10. The operating method of claim 9, further comprising: receiving, from the wireless charging station, the amount of power charged in the vehicle; and transmitting, to the server, the received amount of power in association with the vehicle information.

11. The operating method of claim 7, further comprising: controlling the operation state of the wireless charging station to switch from a sleep state to a standby state when a signal information of traffic lights on the road indicates that the vehicle is to be stopped due to the traffic lights in a traveling direction.

12. The operating method of claim 7, further comprising: controlling the wireless charging station to stop power transmission before a signal for a traveling direction of the vehicle in traffic lights on the road is changed from a stop signal to a go signal.
